SYSDESIGN REFERENCE

Content Delivery Networks

Core concepts, reference diagram, tradeoffs, and interview prompts.

Serve cacheable content near users while preserving origin correctness and invalidation controls.

VERSION 1.0 REVIEW

REVIEWED THROUGH 2026-07-18

STATUS HUMAN REVIEW REQUIRED

SYSDESIGN / CDN

SYSDESIGN

Content Delivery Networks

CORE IDEA

Serve cacheable content near users while preserving origin correctness and invalidation controls.

Reference architecture decision chain

01	02	03	04
Cache Key	Ttl	Purge	Signed Urls

Decisions to defend

+ Cache key
+ TTL
+ purge
+ signed URLs
+ origin shielding
+ and dynamic acceleration.

Failure modes to test

+ Stale or private content leakage
+ cache fragmentation
+ and origin stampede.

Interview flow

- [ ] Clarify users, critical paths, and non-goals
- [ ] Estimate only scale that changes a decision
- [ ] Define API, data model, and source of truth
- [ ] Draw synchronous and asynchronous paths
- [ ] Pressure-test timeout, retry, overload, and recovery
- [ ] Close with tradeoffs and one alternative

Deep-dive drill

Content Delivery Networks

Probe	What a strong answer makes explicit
How does the design behave during a regional dependency failure?	Requirement, decision, cost, failure behavior, and observable metric.
Which metric reveals the first bottleneck?	Requirement, decision, cost, failure behavior, and observable metric.
What consistency does the user actually observe?	Requirement, decision, cost, failure behavior, and observable metric.
How would the design change at ten times the traffic?	Requirement, decision, cost, failure behavior, and observable metric.

Data questions

+ Partition key
+ Versioning
+ Uniqueness
+ Retention
+ Reconciliation

Operations questions

+ SLO
+ Backlog
+ Capacity
+ Alert
+ Failover test

DESIGN DISCIPLINE

For every box: which requirement justifies it, what cost does it introduce, and what happens when it fails?

PUBLISHING STATUS

REVIEW GATE: TECHNICAL REVIEW. AI-produced draft. A named human must verify technical accuracy, sourcing, freshness, and editorial quality before publication. Reviewed-through date: 2026-07-18.